

ECHS MATHEMATICS

IB Diploma Programme - Mathematics: Applications and Interpretation SL

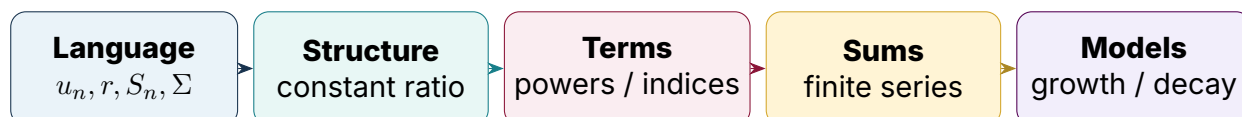
1.3

Geometric Sequences and Series

Full Notes - Guided Examples - IB-Style Tasks - GDC/TI-84 Integration

Central idea. A geometric sequence is a **discrete multiplicative model**: equal changes in the index produce equal **relative** changes in the term value.

$$\boxed{u_{n+1} = ru_n} \implies \boxed{u_n = u_1 r^{n-1}} \implies \boxed{S_n = u_1 \frac{1 - r^n}{1 - r}}$$



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Version: Student Full Notes

Learning route

define → model → calculate → verify → interpret

Learning intentions

By the end of Lesson 1.3, you should be able to:

1. distinguish **additive** change from **multiplicative** change;
2. identify and justify geometric structure using a **constant common ratio**;
3. interpret positive, fractional and negative values of r ;
4. move among **list, recursive, explicit, tabular, graphical, and contextual** representations;
5. find terms, indices, u_1 , and r , including from **separated terms**;
6. use **sigma notation** and finite geometric-series formulas accurately;
7. solve **growth, decay, inverse, and threshold** problems with integer-stage checks;
8. use a GDC transparently, retain guard digits, and evaluate model limitations.

Vocabulary you must use precisely

geometric sequence	se-	a sequence generated by repeated multiplication by the same factor r
common ratio		r , where $u_{n+1} = ru_n$; if $u_n \neq 0$, then $r = \frac{u_{n+1}}{u_n}$
growth factor		a multiplier $r > 1$, e.g. 1.12 means 12% growth per stage
decay factor		a multiplier $0 < r < 1$, e.g. 0.82 means 18% decay per stage
alternating sequence	se-	signs alternate, usually because $r < 0$
geometric series		a sum of geometric-sequence terms
partial sum		$S_n = u_1 + u_2 + \cdots + u_n$
threshold		a target condition such as $u_n > 600$ or $u_n < 10$
discrete model		a model defined at integer stage values, not every real input

Opening problem - community growth**GDC USEFUL**

A community programme receives 80 new members in week 1. Each week, the number of new members is 18% greater than in the previous week.

- A.** How many new members are predicted in week 12?
- B.** How many members join during the first 12 weeks altogether?
- C.** In which week does the weekly prediction first exceed 600?
- D.** Why might this model become unreliable over a long period?

WORKING / ANNOTATIONS

1 - Geometric structure

constant relative change

Arithmetic versus geometric

Arithmetic: 4, 7, 10, 13, ...

constant *difference*: +3

$$\text{Arithmetic: } u_{n+1} - u_n = d$$

Geometric: 4, 12, 36, 108, ...

constant *ratio*: $\times 3$

$$\text{Geometric: } u_{n+1} = ru_n.$$

The defining invariant

A sequence is geometric if one constant value r satisfies

$$u_{n+1} = ru_n.$$

Whenever the preceding term is non-zero,

$$r = \frac{u_{n+1}}{u_n}.$$

To justify a listed sequence, calculate several consecutive ratios. For a formula-defined infinite sequence, show algebraically that $\frac{u_{n+1}}{u_n}$ is constant wherever the quotient is defined.

The sign and magnitude of r

$r > 1$	positive growth; same sign, magnitude increases
$0 < r < 1$	positive decay; same sign, magnitude approaches 0
$r = 1$	constant sequence
$-1 < r < 0$	alternating decay; signs alternate, magnitude decreases
$r = -1$	signs alternate with constant magnitude
$r < -1$	alternating growth; signs alternate, magnitude increases

Worked Example 1

GDC NOT NEEDED

Show that 12, 18, 27, 40.5, ... is geometric and interpret the ratio.

WORKING / ANNOTATIONS

Worked Example 2

GDC NOT NEEDED

Show that $5, -10, 20, -40, \dots$ is geometric and describe its behaviour.

WORKING / ANNOTATIONS

Your Turn 1

GDC NOT NEEDED

Decide whether each sequence is geometric. If so, state r .

A. $3, 12, 48, 192, \dots$

D. $1, 4, 9, 16, \dots$

B. $64, 32, 16, 8, \dots$

E. $5, 5, 5, 5, \dots$

C. $2, -6, 18, -54, \dots$

F. $12, 6, 2, 0.5, \dots$

WORKING / ANNOTATIONS

Percentage language

Repeated growth by $p\%$ means

$$r = 1 + \frac{p}{100}.$$

Repeated decay by $p\%$ means

$$r = 1 - \frac{p}{100}.$$

For example, $r = 0.86$ means 86% remains each stage, so there is a 14% decrease per stage.

Checkpoint - structure and interpretation

1. State the ratio of 18, 54, 162, 486, ...
2. Explain the difference between $r = 1.12$ and $r = 0.12$.
3. Describe the behaviour of a geometric sequence with $r = -0.4$.
4. Decide whether 8, 12, 18, 27, ... is geometric.

WORKING / ANNOTATIONS

2 - Recursive and explicit forms

repeated multiplication becomes a power

Recursive form

$$u_1 = a, \quad u_{n+1} = ru_n$$

This form tells you the starting value and how to generate the *next* term.

Explicit form

$$u_n = u_1 r^{n-1}$$

The exponent is $n - 1$ because term 1 has undergone **zero** multiplications.

$$u_1 = u_1 r^0, \quad u_2 = u_1 r^1, \quad u_3 = u_1 r^2, \quad \dots, \quad u_n = u_1 r^{n-1}.$$

Common trap

Writing $u_n = u_1 r^n$. Always test $n = 1$: your formula must return u_1 .

Worked Example 3

GDC NOT NEEDED

For 10, 20, 40, 80, ..., find u_{10} .

WORKING / ANNOTATIONS

Worked Example 4

GDC NOT NEEDED

For $u_1 = 96$ and $r = -\frac{1}{2}$, find u_7 .

WORKING / ANNOTATIONS

Your Turn 2

GDC NOT NEEDED

Complete each representation.

- A. List $\rightarrow$ explicit: 3, 12, 48, 192, ...
- B. Explicit $\rightarrow$ first four terms: $u_n = 81 \left(\frac{1}{3}\right)^{n-1}$
- C. Recursive $\rightarrow$ explicit: $u_1 = -5$, $u_{n+1} = -2u_n$
- D. Explicit $\rightarrow$ recursive: $u_n = 7(0.8)^{n-1}$

WORKING / ANNOTATIONS

Discrete graph interpretation

A sequence graph consists of **points at integer indices**. A smooth exponential curve through the points can help visualize the pattern, but the curve contains values that are *not* sequence terms.

$$(n, u_n), \quad n = 1, 2, 3, \dots$$

Graph expectations before technology

- $r > 1$: positive points rise rapidly.
- $0 < r < 1$: positive points decrease toward 0.
- $r < 0$: points alternate above and below the horizontal axis.
- $|r| > 1$: magnitudes grow; $|r| < 1$: magnitudes decay.

TI-84 sequence table

GDC SKILL

For

$$u_n = 120(0.86)^{n-1},$$

enter

$$Y_1 = 120(0.86)^{X-1}.$$

Then use **2nd WINDOW (TBLSET)** with TblStart = 1, Δ Tbl = 1, and open **2nd GRAPH (TABLE)**.

Expected first values:

$$120, 103.2, 88.752, 76.32672, 65.6409792, \dots$$

Verify independently:

$$\frac{103.2}{120} = 0.86.$$

IB communication tip

The calculator table supports the representation; the mathematical model is established by the constant ratio or the explicit rule, not by the display alone.

3 - Terms and indices

powers locate distant stages

Finding a distant term

If u_1 and r are known, substitute directly:

$$u_n = u_1 r^{n-1}.$$

For large powers, a GDC is efficient. Retain guard digits until the final rounding step.

Large powers and guard digits

GDC SKILL

Evaluate

$$8500(1.037)^{18}.$$

A TI-84 entry is

$$8500(1.037)^{18},$$

which gives

$$16346.9427149 \dots$$

Do not round 1.037^{18} first. If the context requires two decimal places, report 16346.94 only at the end.

Worked Example 5

GDC NOT NEEDED

The geometric sequence has $u_1 = 7$ and $r = 3$. Find u_8 .

WORKING / ANNOTATIONS

Does a target value occur?

To test whether T is a term, solve

$$u_1 r^{n-1} = T.$$

A real-valued solution for n is not enough. The index must be a valid integer stage.

Worked Example 6**GDC USEFUL**

Does 5000 occur in $u_n = 20(1.5)^{n-1}$?

WORKING / ANNOTATIONS**Your Turn 3****GDC USEFUL**

- A. Find u_{12} when $u_1 = 5$, $r = 1.4$.
- B. Find u_9 when $u_1 = 128$, $r = \frac{1}{2}$.
- C. Does 640 occur in $5, 10, 20, 40, \dots$? If yes, state the index.

WORKING / ANNOTATIONS

4 - Recovering u_1 and r

separated terms create powers of the ratio

Separated-term relation

For terms u_p and u_q ,

$$u_p = u_1 r^{p-1}, \quad u_q = u_1 r^{q-1}.$$

Dividing eliminates u_1 :

$$\frac{u_q}{u_p} = r^{q-p}.$$

The index gap $q - p$ becomes the exponent.

Even-gap ambiguity

If the index gap is even and

$$r^{2k} = c > 0,$$

then both $r = +\sqrt[2k]{c}$ and $r = -\sqrt[2k]{c}$ may be possible. A negative ratio can be hidden because even powers remove its sign.

Worked Example 7

GDC NOT NEEDED

Given $u_2 = 12$ and $u_5 = 324$, find r , u_1 , and u_{10} .

WORKING / ANNOTATIONS

Worked Example 8

GDC NOT NEEDED

Given $u_5 = 3$ and $u_7 = 0.75$, find all possible real values of r .

WORKING / ANNOTATIONS

Your Turn 4

GDC NOT NEEDED

Find every admissible real ratio and the corresponding first term.

- A. $u_3 = 20, u_6 = 160$
- B. $u_2 = 18, u_4 = 2$
- C. $u_4 = -54, u_7 = 1458$

WORKING / ANNOTATIONS

Recover the first term

After finding r ,

$$u_1 = \frac{u_p}{r^{p-1}}.$$

Always verify the recovered u_1 and r against *both* given terms.

Three consecutive geometric terms

If A, B, C are three consecutive non-zero geometric terms, then

$$B^2 = AC.$$

Reason:

$$B = Ar, \quad C = Ar^2 \implies B^2 = A^2 r^2 = AC.$$

Worked Example 9

GDC NOT NEEDED

The first three terms are $x, x + 6, 4x$. Find all real values of x that make them geometric.

WORKING / ANNOTATIONS

Checkpoint - terms and parameters

1. Find u_{12} when $u_1 = 5, r = 1.4$.
2. Find r when $u_3 = 16, u_6 = 128$.
3. Find all r when $u_2 = 12, u_4 = 3$.
4. Find u_1 for each result in question 3.
5. If $4, x, 36$ are positive consecutive terms, find x .
6. Explain why an even index gap can hide sign alternation.

WORKING / ANNOTATIONS**A useful sign check**

When $q - p$ is odd, the sign of r is determined by the sign of u_q/u_p . When $q - p$ is even, a positive ratio u_q/u_p may allow both a positive and negative common ratio.

5 - Growth and decay thresholds

locate continuously, decide discretely

First term above a threshold

For $r > 1$,

$$u_1 r^{n-1} > T.$$

Logs, a solver, or a table can locate the crossing, but the final answer must be the **least integer stage** satisfying the original inequality.

Worked Example 10

GDC USEFUL

A population begins at 2500 and grows by 35% per period. Find the first period for which it exceeds 20000.

WORKING / ANNOTATIONS

The strongest threshold answer

A complete threshold solution has four pieces:

model → inequality → integer decision → adjacent-stage check.

The last check proves that the reported stage is the *first* one satisfying the condition.

Your Turn 5

GDC USEFUL

Find the first valid index and verify adjacent terms.

A. $u_1 = 600, r = 1.25$: first term above 6000.

B. $u_1 = 500, r = 1.22$: first term above 5000.

WORKING / ANNOTATIONS

First term below a threshold

For $0 < r < 1$,

$$u_1 r^{n-1} < T.$$

Because $\log r < 0$, manual logarithmic rearrangement is easy to mishandle. A table or numerical solver is often efficient, followed by an adjacent-stage check.

Worked Example 11

GDC USEFUL

A medication level is 100 mg initially and retains 78% each hour. Find the first recorded term below 10 mg.

WORKING / ANNOTATIONS

Your Turn 6

GDC USEFUL

For $u_1 = 200, r = 0.85$, find the first term below 25. Verify the adjacent stages.

WORKING / ANNOTATIONS

Common trap

A calculator result such as $n = 13.42$ is a *continuous crossing*, not automatically the contextual answer. If n counts discrete stages, test 13 and 14.

TI-84 threshold workflow

calculator evidence must support the mathematics

Example - geometric decay threshold

GDC SKILL

Goal: find the first integer n for which

$$P_n = 15000(0.82)^n < 1000.$$

1. Estimate the crossing with logs

$$\frac{\log(1000/15000)}{\log(0.82)} \approx 13.64594.$$

2. Enter the model

$$Y_1 = 15000(0.82)^X.$$

3. Inspect integer rows near the crossing

$$P_{13} \approx 1136.77, \quad P_{14} \approx 932.15.$$

4. Conclude

$$1136.77 \geq 1000 > 932.15,$$

so the first valid discrete stage is 14.

What to write in an IB-style solution

Do not write only "the calculator gave 14." Write the **model**, the **continuous estimate or table method**, and the **adjacent values** that prove minimality.

GDC decision drill

For each task, state whether the calculator mainly helps with *generation*, *large powers*, *finite sums*, or *threshold search*.

1. $u_n = 120(0.86)^{n-1}$: list the first five terms.
2. $8500(1.037)^{18}$.
3. $\sum_{k=0}^{24} 600(1.025)^k$.
4. First n with $15000(0.82)^n < 1000$.

WORKING / ANNOTATIONS

6 - Finite geometric series

accumulate repeated multiplicative effects

Sequence term versus partial sum

$u_n = u_1 r^{n-1}$ is one term, while $S_n = u_1 + u_2 + \cdots + u_n$ is the cumulative total of the first n terms.

Sigma notation

$$\sum_{k=1}^8 5(2)^{k-1}$$

means

$$5 + 10 + 20 + \cdots + 640.$$

The lower limit tells you the first index; the upper limit tells you the last index; the summand tells you the term formula.

Inclusive term count

For integer indices from p to q ,

$$\boxed{\# \text{ terms} = q - p + 1}.$$

Thus $k = 0, \dots, 24$ contains 25 terms, not 24.

TI-84 sigma sum starting at index zero

GDC SKILL

Evaluate

$$\sum_{k=0}^{24} 600(1.025)^k.$$

Use

$$\text{sum(seq}(600(1.025)^X, X, 0, 24))$$

or the equivalent TI-84 list commands. The result is

$$20494.6583597 \dots$$

The $k = 0$ term is 600. Starting at $k = 1$ would incorrectly omit it.

IB communication tip

Index bounds are mathematical information, not formatting details. Always count the terms and identify the first included term before pressing Enter.

7 - Deriving the finite-sum formula

multiply, align, subtract

Derivation

Let

$$S_n = u_1 + u_1r + u_1r^2 + \cdots + u_1r^{n-1}.$$

Multiply by r :

$$rS_n = u_1r + u_1r^2 + \cdots + u_1r^n.$$

Subtract:

$$S_n - rS_n = u_1 - u_1r^n.$$

Therefore

$$(1 - r)S_n = u_1(1 - r^n),$$

and, for $r \neq 1$,

$$S_n = u_1 \frac{1 - r^n}{1 - r}.$$

Equivalent form

Multiplying numerator and denominator by -1 gives

$$S_n = u_1 \frac{r^n - 1}{r - 1}.$$

Both forms are equivalent. Choose the form that reduces sign errors.

Edge case: r

If $r = 1$, the sequence is constant:

$$u_1, u_1, u_1, \dots$$

so

$$S_n = nu_1.$$

The fraction formula is not used because its denominator would be 0.

Worked Example 12

GDC NOT NEEDED

Calculate $2 + 4 + 8 + \cdots + 1024$.

WORKING / ANNOTATIONS

Worked Example 13**GDC NOT NEEDED**

Find the sum of the first 8 terms of 64, 32, 16, ...

WORKING / ANNOTATIONS**Worked Example 14****GDC NOT NEEDED**

Calculate $12 - 6 + 3 - 1.5 + 0.75$.

WORKING / ANNOTATIONS**Your Turn 7****GDC NOT NEEDED**

Calculate each finite sum.

- A. First 7 terms of 3, 9, 27, ...
- B. First 6 terms of 160, 80, 40, ...
- C. $5 - 10 + 20 - 40 + 80$
- D. $\sum_{k=3}^8 2(3)^{k-1}$

WORKING / ANNOTATIONS

Checkpoint - finite geometric series

1. Expand $\sum_{k=1}^4 3(2)^{k-1}$.
2. How many terms are in $\sum_{k=7}^{25} u_k$?
3. Find S_{10} when $u_1 = 5, r = 1.2$.
4. Find S_6 when $u_1 = 32, r = -\frac{1}{2}$.
5. State the sum formula when $r = 1$.
6. Explain why term count must be known before using the finite-sum formula.

WORKING / ANNOTATIONS

Shifted geometric block

A block

$$\sum_{k=p}^q u_k$$

is itself a geometric series with first included term u_p , common ratio r , and

$$q - p + 1$$

terms. Hence

$$\sum_{k=p}^q u_k = u_p \frac{1 - r^{q-p+1}}{1 - r}.$$

Worked Example 15**GDC NOT NEEDED**

For $u_1 = 4, r = 2$, calculate $u_5 + u_6 + \cdots + u_{10}$.

WORKING / ANNOTATIONS

8 - Inverse finite-series problems

the unknown may be the term count

Solve for the term count n

If

$$S_n = T,$$

then

$$u_1 \frac{1 - r^n}{1 - r} = T.$$

Rearrange to isolate r^n , then use exact powers, logarithms, or a solver. Keep only admissible positive integer values of n .

Worked Example 16

GDC NOT NEEDED

For $3 + 6 + 12 + \dots$, find n if $S_n = 3069$.

WORKING / ANNOTATIONS

Your Turn 8

GDC USEFUL

For $u_1 = 5, r = 1.3$, find the least positive integer n for which $S_n > 500$. Verify your answer by comparing two adjacent partial sums.

WORKING / ANNOTATIONS

Inverse-problem discipline

If the algebra produces a non-integer n but n counts whole terms, rows, deposits, or recorded stages, the final response must respect the discrete domain.

9 - Growth and decay modelling

translate percentage change into a multiplier

Repeated growth

If a quantity increases by $p\%$ each equal stage,

$$r = 1 + \frac{p}{100}.$$

Example: 12% growth gives $r = 1.12$.

Repeated decay

If a quantity decreases by $p\%$ each equal stage,

$$r = 1 - \frac{p}{100}.$$

Example: 18% decay means 82% remains, so $r = 0.82$.

Term value or cumulative total?

Before choosing a formula, translate the words.

Per-period value

members joining in week n

medicine remaining at reading n

files produced in cycle n

Cumulative total

total members joining through week n

total of recorded amounts across n readings

total files produced in the first n cycles

Worked Example 17

GDC USEFUL

A bacteria culture has 450 cells at stage 1 and grows by 22% per stage. Find the stage-9 value and the total of the first 9 recorded stage values.

WORKING / ANNOTATIONS

Your Turn 9

GDC USEFUL

A quantity begins at 120 and retains 86% each stage.

- A. State the percentage decrease.
- B. Find u_6 .
- C. Find S_{12} .
- D. Explain what S_{12} means and when it may *not* be physically meaningful.

WORKING / ANNOTATIONS

10 - Model evaluation

a good calculation can still come from a poor model

Questions to ask before trusting a geometric model

1. Are stages equally spaced in time or another meaningful variable?
2. Is the same percentage change plausible at every stage?
3. Is there a natural maximum, minimum, capacity, or saturation level?
4. Are the outputs continuous measurements or discrete counts?
5. Does rounding to whole people/items change future values?
6. Is the model being used only within the data range where it was justified?

Discrete versus continuous interpretation

A smooth curve $y = u_1 r^{x-1}$ can be useful for locating a crossing, but a sequence contains only the intended integer stages. Use the continuous graph to *locate*; use the discrete model to *decide*.

Calculator decision rule

GDC not needed: simple ratio identification, short exact terms, exact sign reasoning, or a short proof.

GDC useful: large powers, many-term sums, logarithmic thresholds, numerical searches, or verification across many stages.

Always: define the mathematics first and independently verify a key result when practical.

Your Turn 10

GDC NOT NEEDED

A social-media account is modeled as growing by 40% every week forever. Give **two** reasons this may be a poor long-term model even if the first few weeks fit well.

WORKING / ANNOTATIONS

11 - Optional extension

beyond the core finite-series work

Infinite geometric series

EXTENSION

This preview is useful for mathematical connection, but the central Lesson 1.3 assessment focus is the **finite** geometric series.

If $|r| < 1$, then $r^n \rightarrow 0$ as $n \rightarrow \infty$. Starting from

$$S_n = u_1 \frac{1 - r^n}{1 - r},$$

the limiting sum is

$$S_\infty = \frac{u_1}{1 - r}.$$

If $|r| \geq 1$, the terms do not approach 0 in the required way, so the infinite geometric series does not converge to a finite sum.

Worked Example 18

EXTENSION

Find the infinite sum $12 + 6 + 3 + 1.5 + \dots$.

WORKING / ANNOTATIONS

Your Turn 11

EXTENSION

Decide whether each infinite series converges. If it does, find the sum.

A. $20 + 10 + 5 + \dots$

B. $3 - 6 + 12 - 24 + \dots$

WORKING / ANNOTATIONS

12 - Synthesis

one connected multiplicative structure

Core formula sheet

Geometric recurrence: $u_{n+1} = ru_n$,

Ratio test: $r = \frac{u_{n+1}}{u_n} \quad (u_n \neq 0)$,

Explicit term: $u_n = u_1 r^{n-1}$,

Separated terms: $\frac{u_q}{u_p} = r^{q-p}$,

Consecutive-term relation: $B^2 = AC$,

Finite sum: $S_n = u_1 \frac{1 - r^n}{1 - r} \quad (r \neq 1)$,

Equivalent sum: $S_n = u_1 \frac{r^n - 1}{r - 1}$,

Constant case: $r = 1 \Rightarrow S_n = nu_1$,

Shifted count: # terms from p to $q = q - p + 1$.

Problem-selection map

Question asks for one distant term

use $u_n = u_1 r^{n-1}$

Two separated terms given

divide: $u_q/u_p = r^{q-p}$

Three consecutive terms

use $B^2 = AC$ when efficient

Sum of first n terms

use finite geometric-sum formula

Sum from index p to q

rebase at u_p , count $q - p + 1$

First stage above/below a target

solve approximately, then verify adjacent integers

Repeated percentage change

convert percentage to a factor first

Four questions students must separate

1. What is the value of term n ? $\rightarrow u_n$
2. What is the total of the first n terms? $\rightarrow S_n$
3. At which integer stage is a threshold first crossed? $\rightarrow$ solve + adjacent check
4. Does the model remain believable? $\rightarrow$ evaluate assumptions and context

13 - Mixed practice A

structure, terms, parameters

Mixed practice

1. State the common ratio of $162, 54, 18, 6, \dots$
2. Find u_7 when $u_1 = -4, r = -3$.
3. Find n if $5(2)^{n-1} = 640$.
4. Given $u_2 = 12, u_5 = 324$, find u_1 and r .
5. Given $u_3 = 8, u_5 = 2$, find all real values of r and the corresponding u_1 .
6. Explain why $r = -0.6$ produces an alternating decay sequence.

WORKING / ANNOTATIONS

14 - Mixed practice B

sums, sigma, thresholds, interpretation

Mixed practice

1. Find the first 8-term sum for $u_1 = 2, r = 1.5$.
2. Evaluate $\sum_{k=2}^6 5(2)^{k-1}$.
3. Find the first n for which $900(1.12)^{n-1} > 2000$, and verify adjacent stages.
4. Find the first n for which $500(0.82)^{n-1} < 50$, and verify adjacent stages.
5. Interpret $r = 0.92$ as a percentage change.
6. Explain why the real solution $n = 8.31$ is not automatically the answer to a question asking for the first whole stage above a target.

WORKING / ANNOTATIONS

15 - IB-style task 1

growth model and communication

IB-style extended response - Community programme growth**13 marks****GDC USEFUL**

The weekly numbers of new members joining a community programme are modeled by a geometric sequence with first term 80 and common ratio 1.18.

- (a) Interpret the common ratio in context. [2]
- (b) Find the predicted number joining in week 12. [2]
- (c) Calculate the total predicted number joining in the first 12 weeks. [3]
- (d) Determine the first week in which the weekly prediction exceeds 600. Verify that it is the first. [3]
- (e) State how the whole-person context should affect reporting. [1]
- (f) State one limitation of the geometric model. [2]

WORKING / ANNOTATIONS

16 - IB-style task 2

sign ambiguity and finite sum

IB-style extended response - Hidden sign alternation**12 marks**

A geometric sequence satisfies

$$u_2 = 18, \quad u_4 = 2.$$

- (a) Show that $r^2 = \frac{1}{9}$. [2]
- (b) Hence state the two possible real values of r . [2]
- (c) For each value of r , find the corresponding u_1 . [2]
- (d) Write the first five terms for the negative-ratio sequence. [2]
- (e) Calculate the sum of the first five terms of the negative-ratio sequence. [3]
- (f) Explain why the two possible sequences have the same second and fourth terms. [1]

WORKING / ANNOTATIONS

17 - GDC mastery check

model first, enter carefully, verify independently

Calculator-aware questions

- Which TI-84 table entry correctly represents $u_n = 120(0.86)^{n-1}$?
 (A) $120(0.86)^X$
 (B) $120(0.86)^{X-1}$
 (C) $0.86(120)^{X-1}$
 (D) $120 - 0.86(X - 1)$
- Why should $\text{TblStart} = 1$ and $\Delta\text{Tbl} = 1$ usually be used for this sequence?
- How many terms are in $\sum_{k=0}^{24} 600(1.025)^k$?
- What is wrong with summing from $k = 1$ to 24 instead?
- A solver returns $n = 13.42$ for a discrete threshold. What must happen next?
- Why is keeping guard digits important in $8500(1.037)^{18}$?

WORKING / ANNOTATIONS

Calculator principle

Technology can **generate, evaluate, search, and verify**. It does not replace the need to define the geometric model, interpret the index, or justify the final integer decision.

18 - Final exit ticket

can you separate term, sum, stage, and model?

Exit ticket

For

$$u_1 = 40, \quad r = 1.15,$$

answer the following.

1. Write the explicit formula for u_n .
2. Find u_{10} .
3. Find S_{10} .
4. Find the first n for which $u_n > 150$.
5. Verify your threshold answer using adjacent terms.
6. State one realistic context in which this model could be useful for a limited number of stages.
7. State one reason the model might fail in the long run.

WORKING / ANNOTATIONS

Reflection

Complete these before leaving:

- The most important difference between arithmetic and geometric structure is _____.
- The formula I must not confuse with a term formula is _____.
- When a calculator gives a decimal stage number, I should _____.
- One modelling limitation I will look for is _____.

Next lesson

Lesson 1.4 applies repeated multiplicative change to financial contexts. The same ideas - percentage factors, powers, finite sums, indexing, and transparent technology use - become financial mathematics.